

# Leaf Stomata Microscope Detective

Count the breathing pores and rank leaves by how they manage water.

**Win condition:** Most accurate detective ranking with clean data and strong evidence.

## THE SCIENCE YOU ARE USING

**Stomata: pores for gas exchange.** Leaves breathe through tiny pores called stomata. They open to take in carbon dioxide and let oxygen out, but open stomata also lose water. The number and spacing of stomata reflect how a plant balances feeding itself against drying out.

**Sampling and counting.** You make a safe leaf peel or use prepared slides, count stomata in several fields of view, and average. Comparing counts across leaves lets you rank them by water strategy with real data.

## HOW TO BUILD AND RUN IT

- 01 Make a safe peel.** Paint clear polish on the leaf underside, let it dry, and lift the print with clear tape. Or use a prepared slide.
- 02 Focus the field.** Place the peel under the microscope and focus until the stomata are sharp.
- 03 Count several fields.** Count stomata in three or more fields of view and average to reduce error.
- 04 Compare leaves.** Repeat for different leaves and build a team data table of counts per field.
- 05 Rank by water strategy.** Use your counts to rank leaves from water-saving to water-spending, citing the data.

**Safety:** Use polish only at ventilated instructor station. No chromatography, no solvents, no tie-dye. Allergy: Plant sensitivity note. Prepared slides available.

## MATERIALS

|                               |             |
|-------------------------------|-------------|
| Leaf samples                  | varied      |
| Clear tape                    | shared      |
| Clear nail polish or slides   | shared      |
| USB microscopes or magnifiers | per team    |
| Grid sheets                   | per team    |
| Gloves                        | per student |

**YOUR PLAN AND DATA**

Clean, complete data is worth as much as a fast result.

**Prediction (what will work best, and why):** \_\_\_\_\_

**The ONE variable we are testing:** \_\_\_\_\_

**Baseline result:** \_\_\_\_\_

**After redesign:** \_\_\_\_\_

**DEFEND YOUR DESIGN**

**What evidence made you change your design?** \_\_\_\_\_

**If you ran it again, what is the ONE thing you would change?** \_\_\_\_\_

**HOW YOU SCORE (100 POINTS)**

| SCORING CRITERION     | POINTS     |
|-----------------------|------------|
| Slide or peel quality | 25         |
| Counting accuracy     | 30         |
| Ranking evidence      | 25         |
| Team data table       | 10         |
| Cleanup and care      | 10         |
| <b>Total</b>          | <b>100</b> |